

THEORY CONCEPTS & DEFINITIONS

Holistic Data Preparer — Final Project

A plain-language reference explaining every core concept and technique used across the Customer Credit Risk data preprocessing and feature engineering project (Parts A–H).

*Prepared for: Holistic Data Preparer (Final Project)
Red & White Skill Education*

1 Data Analysis

The process of inspecting, cleaning, transforming and modeling data to discover useful information, reach conclusions, and support decision-making. It combines statistics, visualization and domain knowledge to turn raw data into actionable insight.

2 Planning a Data Science Project

A structured project generally follows the CRISP-DM style stages:

- **Business Understanding** — define the problem in business terms.
- **Data Understanding** — identify sources, sample and assess quality.
- **Data Preparation** — clean, encode, scale and engineer features.
- **Modeling** — train candidate ML algorithms.
- **Evaluation** — validate against the original business objective.
- **Deployment** — ship and monitor the model in production.

3 Framing a Machine Learning Problem

Translating a business question into a solvable ML task by defining the prediction objective, problem type (classification / regression / etc.), target variable, input features, evaluation metric, and constraints (e.g. interpretability, fairness, latency).

4 Tensors

A tensor is a generalization of scalars, vectors and matrices to n dimensions — the core data structure of ML frameworks (NumPy, PyTorch, TensorFlow).

Rank	Name	Example	NumPy shape
0	Scalar	a single number, e.g. 5	()
1	Vector	one customer's feature row	(n_features,)
2	Matrix	the whole dataset table	(n_rows, n_features)
3+	Tensor	a batch of records over time	(batch, rows, features)

5 Missing Value Handling Techniques

■ Simple Imputer

replaces missing values with a fixed statistic: mean/median for numeric columns, most-frequent for categorical columns.

■ Most Frequent Category Imputation

fills categorical gaps with the mode of the column.

■ Missing Indicator + Random Sample Imputation

adds a binary flag marking where data was missing (since missingness can itself be predictive), then fills gaps by sampling from the observed distribution to preserve its shape.

■ KNN Imputer

estimates a missing value from the average of its k nearest neighbours across other features, capturing multivariate relationships.

■ MICE (Multiple Imputation by Chained Equations)

iteratively models each feature with missing values as a function of the other features, cycling until estimates stabilize.

■ Complete Case Analysis

simply drops rows or columns containing missing values; simplest but can discard large amounts of valid data.

6

Outlier Detection & Treatment

■ Z-score Method

flags a value as an outlier if it is more than a chosen number of standard deviations from the mean (commonly $|z| > 3$).

■ IQR Method

flags values below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$, where $IQR = Q3 - Q1$.

■ Percentile Method

flags values below/above a fixed percentile cutoff (e.g. 1st/99th).

■ Winsorization

instead of removing outliers, caps extreme values at a chosen percentile, preserving sample size while limiting their influence.

7

Variable Types & Date Handling

Mixed variables combine numeric and categorical fields within the same record and must be handled with type-appropriate techniques. Date/time variables are decomposed into components — Year, Month, Day, Weekday — so models can learn seasonality and tenure effects instead of treating a date as an opaque string.

8

Categorical Encoding

■ Ordinal Encoding

maps ordered categories (e.g. Primary < Secondary < Graduate) to increasing integers, preserving their natural rank.

■ Label Encoding

assigns an arbitrary integer to each category; best for binary or tree-based-model-friendly categorical fields.

■ One-Hot Encoding

creates a separate binary column per category, avoiding any false sense of order for nominal variables like region or loan purpose.

9 Numerical Feature Encoding

■ Binning / Discretization

groups a continuous variable into labeled ranges (e.g. income tiers).

■ Binarization

converts a numeric column into 0/1 based on a threshold.

■ Quantile Binning

splits data into equal-frequency buckets (e.g. quartiles).

■ K-Means Binning

uses 1-D K-Means clustering to find natural groupings in a numeric feature.

10 Feature Scaling

■ Standardization (Z-score scaling)

rescales a feature to zero mean and unit variance.

■ Normalization

rescales each sample (row) to have unit norm.

■ Min-Max Scaling

rescales values into a fixed range, typically [0, 1].

■ MaxAbs Scaling

divides by the maximum absolute value, preserving sparsity and sign.

■ Robust Scaling

uses the median and IQR instead of mean/std, making it resistant to outliers.

11 Transformations

■ FunctionTransformer

applies an arbitrary function (log, reciprocal, square root, etc.) to a column, typically to reduce skew.

■ PowerTransformer (Box-Cox / Yeo-Johnson)

statistically optimal power transforms that make skewed data more Gaussian-like; Box-Cox requires strictly positive values, Yeo-Johnson supports zero and negative values too.

■ ColumnTransformer

applies different preprocessing pipelines to different column subsets (e.g. scaling numeric columns while one-hot encoding categorical columns) within a single, reusable pipeline object.

12 Feature Construction

Building new, more informative features from existing raw columns — e.g. Debt-to-Income ratio ($\text{loan_amount} / \text{annual_income}$), Average Monthly Transactions, and Spending-to-Income ratio — often improves model performance more than any single preprocessing step, because it encodes domain knowledge directly into the feature space.

Quick recap: Sections 5–11 above are exactly the technique families applied, in order, inside **Part H** of the notebook to build **final_cleaned_dataset.csv** and **scaled_dataset.csv**.